

AIWARYA BABU

SENIOR SOFTWARE DEVELOPER

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PROFESSIONAL SUMMARY

Senior Software Developer with **2+ years** of experience building **scalable backend systems, multi-tenant SaaS platforms, distributed microservices**, and **AI-driven applications** using **Python, FastAPI, TypeScript, React, and Next.js**. Experienced in **algorithmic trading infrastructure**, real-time communication, **LLM-powered evaluation workflows**, exam proctoring, **payment integrations**, and end-to-end **full-stack development**.

TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic v2), TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3/SASS, Bash

Backend & APIs: FastAPI, SQLAlchemy, SQLAlchemy 2.0, Node.js, REST APIs, WebSockets, Alembic, APScheduler, Playwright

Frontend: React 18/19, Next.js 16 (App Router), Vite, Redux Toolkit, Redux-Saga, Tailwind CSS, Material UI, PrimeReact, React Flow

GenAI & LLM: OpenAI API (GPT-4o), Anthropic Claude SDK, Google Gemini API, Prompt Engineering, Context Injection / RAG, Structured JSON Output, Token & Cost Tracking

Databases & Caching: PostgreSQL, MongoDB (Motor/Async), Redis (Cache & Pub/Sub), SQLite

Data & Media: Pandas, NumPy, OpenPyXL, Apache ECharts, Recharts, MoviePy, PDFPlumber, ReportLab

Integrations: Stripe API & Webhooks, Fyers Trading API, Google Drive API, Firebase Admin, Twilio, Resend, Xero Node SDK

DevOps & Quality: Docker, GitLab CI/CD, SonarQube, Security Scanning

Architecture & Concepts: Microservices, Multi-Tenant SaaS, RBAC, REST API Design, Real-Time Systems, Pub/Sub Messaging, SSR/SSG, Algorithmic Trading, LLM Application Development

PROFESSIONAL EXPERIENCE

Software Developer / Senior Software Developer

February 2024 – Present

Aagnia Technologies | Coimbatore, Tamil Nadu

Full-stack and backend developer across **Everest Tutoring** (AI-powered EdTech) and **Invetaa** (algorithmic trading microservices), covering **backend architecture, APIs**, real-time services, frontend workflows, and third-party integrations. Applied **GitLab CI/CD, Docker, SonarQube**, and security scanning practices across development workflows.

Everest Tutoring – AI-Powered Online Examination & Learning Ecosystem

Tech: Python, FastAPI, SQLAlchemy, PostgreSQL, React 18/19, Next.js 16, TypeScript, OpenAI GPT-4o, Anthropic Claude, Redux Toolkit, Redux-Saga, Tailwind CSS, Material UI, Playwright, VdoCipher API, Stripe, Docker

- Engineered an automated **rubric-based writing evaluation system** using **GPT-4o** with **structured JSON output** and retry/fallback handling.
- Built centralized **LLM token and cost accounting** for prompt tokens, completion tokens, model usage, USD expenditure, and agent routing.
- Created “Elliot,” a **syllabus-aware academic chatbot** with **dynamic context injection** from student performance, weak topics, schedules, and pending learning activities.
- Implemented **VTT/SRT transcript processing** to generate structured lesson notes, summaries, key concepts, and **interactive video quizzes**.
- Integrated **rrweb session recording/replay**, MathLive, KaTeX, and Fabric.js into the examination experience to support **proctoring and audit trails**.
- Built **bulk data ingestion and normalization** with **Pandas** and **OpenPyXL** for questions and video metadata.
- Developed **Everest_Meet_Bot** with **Playwright** for Google Meet joining, session persistence, and automated browser permissions.
- Implemented **subdomain-based multi-tenant isolation** and **role-based access control (RBAC)**, enabling multiple institutions to operate within a shared platform.

- Integrated **Stripe subscription tiers and webhooks**, **VdoCipher API** for secure video delivery, Google Drive resource management, Twilio/Resend messaging, and booklet print-request workflows.

Invetaa – Algorithmic & Quantitative Trading Brokerage Platform

Tech: Python, FastAPI, MongoDB (Motor), Redis, Pandas, NumPy, Fyers REST API & WebSockets, React, TypeScript, Redux-Saga, React Flow (react), Apache ECharts

- Designed a **distributed 4-microservice trading platform**: Algo Engine, **Order Management System (OMS)**, Backtesting Service, and Webhook Consumer, built for **high-throughput** trading workflows.
- Built **quantitative strategy execution**, including **SMA Crossover** workflows for NSE/NFO Futures & Options with **concurrent processing**.
- Developed the OMS for **automated order routing**, Stop-Loss, Take-Profit, Trailing Stop-Loss, and **margin validation** through the **Fyers API**.
- Built a **vectorized historical backtesting engine** with **Pandas and NumPy** for P&L and risk analytics.
- Implemented **Redis Pub/Sub** as the inter-service communication layer, enabling **real-time** market and execution events to flow between microservices, with Fyers streaming/webhooks feeding live broker and market updates.
- Developed a **drag-and-drop visual strategy builder** with **React and React Flow**, plus real-time financial charts and performance visualizations using **Apache ECharts** and Recharts.

Everest Consultation & Booking Platform

Tech: Next.js 16, React 19, TypeScript, Tailwind CSS, Prisma, Stripe

- Built a **Next.js App Router** booking platform for trial bookings and tutor consultations using **SSR/SSG** patterns.
- Integrated **Xero Node SDK** and **Stripe** for invoicing, payment reconciliation, and bookkeeping workflows.

EDUCATION

Bachelor of Commerce in Information Technology (B.Com IT)

2020 – 2023

VLB Janakiammal College of Arts and Science, Coimbatore | CGPA: 7.4 / 10

CERTIFICATIONS

Full Stack Development (MERN Stack) – NSchool Academy

2024

LANGUAGES

English (Professional) • Malayalam (Native) • Tamil (Fluent)